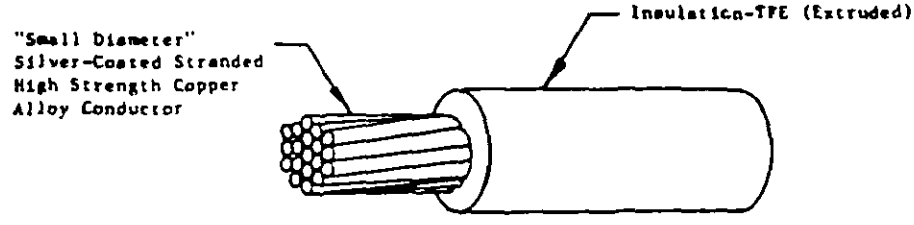
MILITARY SPECIFICATION SHEET

WIRE, ELECTRIC, FLUOROPOLYMER-INSULATED, EXTRUDED TFE,
SILVER-COATED HIGH STRENGTH COPPER ALLOY CONDUCTOR, 600-VOLT

This specification is approved by the Naval Air Systems Command, Department of the Navy.

The complete requirements for procuring the wire described herein shall consist of this document and the issue in *effect of* Specification MIL-W-22759.




TFE-Polytetrafluoroethylene

TABLE I. CONSTRUCTION DETAILS

| Part No. 1/ | Wire size | Stranding (Number of strands X AWG gage of strands) | Diameter of stranded conductor (inches) | | Finished wire | | |
|---|---|---|---|---|---|---|---|
| | | | (min) | (max) | Resistance at 20°C (68°F) (ohms/1000 ft) (max) | Diameter (inches) | Weight (lbs/1000 ft) (max) |
| M22759/22-26-* | 28 | 7 X 36 | .014 | 0.015 | 74.4 | .033 ±.002 | 1.32 |
| M22759/22-26-* | 26 | 19 X 38 | .018 | 0.020 | 44.8 | .038 ±.002 | 1.91 |
| M22759/22-24-* | 24 | 19 X 36 | .023 | 0.024 | 28.4 | .043 ±.002 | 2.61 |
| M22759/22-22-* | 22 | 19 X 34 | .029 | 0.031 | 17.5 | .049 ±.002 | 3.68 |
| M22759/22-20-* | 20 | 19 X 32 | .037 | 0.039 | 10.7 | .058 ±.002 | 5.38 |

1/ PART NO.: The asterisks in the part number column, Tables I and II, shall be replaced by color code designators in accordance with MIL-STD-681. Examples: Size 20, white - M22759/22-20-9; white with orange stripe - M22759/22-20-93.



FSC 61-5

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TABLE II. PERFORMANCE DETAILS

| Part No. | Abrasion resistance (Procedure II) | | | | Bend testing | | | |
|---|---|---|---|---|---|---|---|---|
| | | | | | Mandrel diameter (inches) (±32) | | Test load (lbs) (±32) | |
| | Resistance (inches of tape) (min) (initial condition) | Weight support bracket | Weight (lbs) | Tension load (lbs) | Life cycle (oven & bend tests) 1/ | Cold bend test | Life cycle (oven & bend tests) 1/ | Cold bend test |
| M22759/22-28-* | 12 | A | .50 | 1.0 | .125 | .250 | .50 | |
| M22759/22-26-* | 18 | A | .50 | 1.0 | .125 | .250 | .50 | |
| M22759/22-24-* | 18 | A | .50 | 1.0 | .125 | .250 | .50 | |
| M22759/22-22-* | 24 | A | .50 | 1.0 | .250 | .375 | .75 | |
| M22759/22-20-* | 24 | A | .50 | 1.0 | .250 | .375 | .75 | |

1/ Also for bend tests after immersion.

WIRE RATINGS AND ADDITIONAL REQUIREMENTS

TEMPERATURE RATING: 200°C (392°F) max conductor temperature
VOLTAGE RATING: 600 volts (rms) at sea level
ABRASION RESISTANCE AFTER IMMERSION: No requirement
ACID RESISTANCE: Required
Dielectric test, 2200 volts (rms), 60Hz
BLOCKING: 240 ±2°C (500 ±3.6°F)
COLOR: In accordance with MIL-STD-104, Class 1; white preferred
COLOR STRIPING OR BANDING DURABILITY: 250 cycles (500 strokes) (min), 500 grams weight
DIELECTRIC TEST AFTER IMMERSION: 2200 volts (rms), 60Hz
FLAMMABILITY: Post-flame dielectric test not required
HUMIDITY RESISTANCE: No requirement

IDENTIFICATION DURABILITY: 125 cycles (250 strokes) (min), 500 grams weight
IMPULSE DIELECTRIC TEST: 8.0 kilovolts (peak), IEEE test
INSULATION RESISTANCE: 50,000 megohms for 1000 ft (min)
LIFE CYCLE: Oven temperature 275 ±2°C (527 ±3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
LOW TEMPERATURE (COLD BEND):
Bend temperature: -65 ±2°C (-85 ±3.6°F)
Dielectric test, 2200 volts (rms), 60Hz
SHRINKAGE: 0.03 inch max at 290 ±2°C (554 ±3.6°F)
SMOKE: 290°C (554°F)
SPARK TEST OF PRIMARY INSULATION: Not required
SURFACE RESISTANCE: 500 megohm-inches (min), initial and final readings
THERMAL SHOCK: Oven temperature, 200 ±2°C (392 ±3.6°F)
Max change in measurement, 0.060 inch
WICKING: No requirement
WIRE LENGTH REQUIREMENTS: Schedule A
WRAP TEST: "Wrap back" test required; no cracking
Oven temperature, 111 ±2°C (349.4 ±3.6°F)

Preparing activity:
Navy - AS
(Project No. 6145-N742)



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a. Paragraph Number and Wording:

b. Recommended Wording:

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